

Numerical variables. Frequency distribution table

Dataset	Frequency
1	1
9	1
22	1
24	1
32	1
33	1
41	1
44	1
57	1
66	1
70	1
73	1
75	1
76	1
79	1
82	1
87	1
89	1
95	1
100	1
Total	20

Frequency distribution table				
Desired intervals		5		
Interval width		20		
Interval start	Interval end	Frequency	Relative frequency	
1	21	2	0.10	
21	41	4	0.20	
41	61	3	0.15	
61	81	6	0.30	
81	101	5	0.25	

Numerical variables. Frequency distribution table

Background You are given a dataset.

Task 1 Given that we want to divide the numbers into 6 intervals of equal width, calculate that interval width. Round up to the nearest whole number, bigger than the result that you obtain

Task 2 Create a frequency distribution table that shows

1. The intervals
2. The absolute frequency of each interval
3. The relative frequency of each interval

Task 3 Repeat task 1 and 2, but this time, use the exact interval width. Don't round up to the nearest whole number, that is.

Solution:

Tasks 1 and 2

Dataset	Frequency distribution table			
8				
30	Desired intervals	6		
30	Interval width	46		
50				
86				
94	Interval start	Interval end	Absolute frequency	Relative frequency
102	8	54	4	0.20
110	54	100	2	0.10
169	100	146	2	0.10
170	146	192	3	0.15
176	192	238	1	0.05
236	238	284	8	0.40
240			20	1.00
241				
242				
255	Usually intervals are denoted in this way:			
262	[8,54]	Parentheses () denote that the number is not included		
276	(54,100]	Square brackets [] denote that a number is included		
279	(100,146]	e.g. (100,146] means: from 100 EXCLUDED to 146 INCLUDED		
282	(146,192]			
	(192,238]	The first interval always includes the first number, as in [8,54]		
	(238,284]	the last interval always includes the last number as in (238,284]		

In this lesson and exercise, we have separated them into two columns, for easier manipulation in excel formulas.

Task 3

Frequency distribution table			
Desired intervals		6	
Interval width		45.67	
Interval start	Interval end	Absolute frequency	Relative frequency
8.00	53.67	4	0.20
53.67	99.33	2	0.10
99.33	145.00	2	0.10
145.00	190.67	3	0.15
190.67	236.33	1	0.05
236.33	282.00	8	0.40
		20	1.00

Note that in this way, our intervals start exactly at the first number and finish exactly at the last number from the dataset. This is a fairer representation of reality and that is what you will usually do when doing statistics. However, the corresponding frequencies are the same. The difference between the intervals width is so marginal that it will rarely pose a problem on your analysis. Nevertheless, the norm is to use the exact width.

The histogram

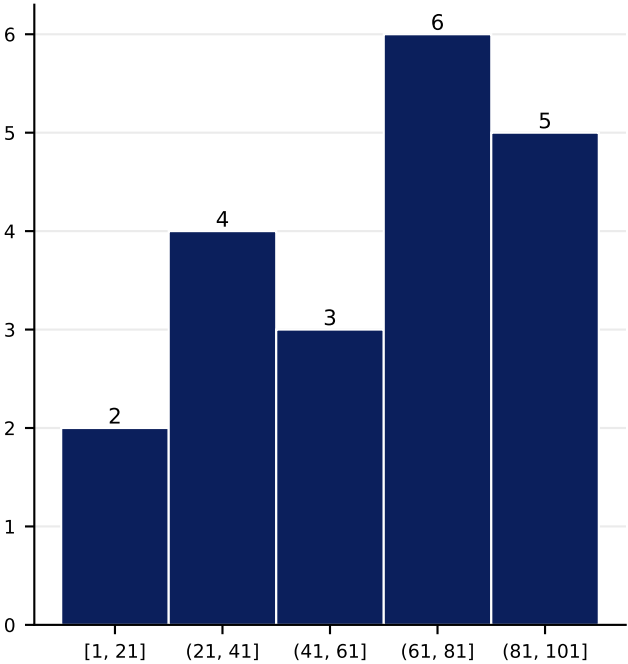
Dataset

1
9
22
24
32
33
42
44
57
66
70
73
75
76
79
82
87
89
95
100

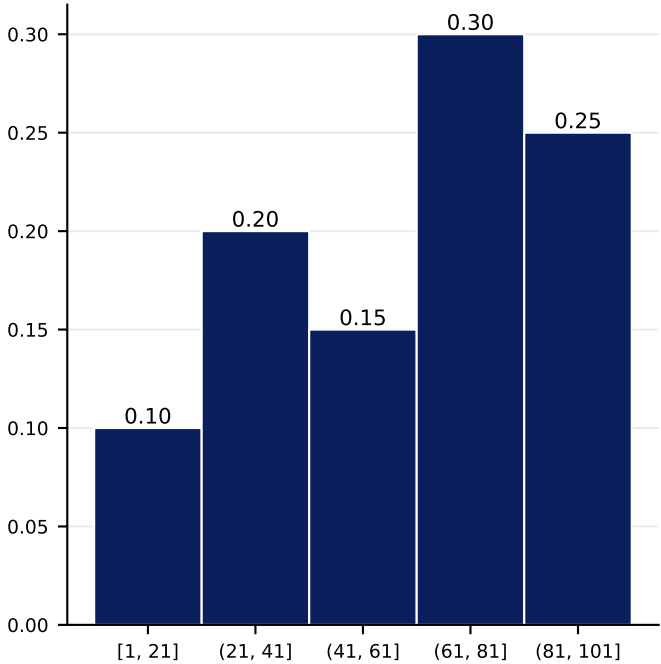
Frequency distribution table

Interval start	Interval end	Frequency	Relative frequency
1	21	2	0.10
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61	81	6	0.30
81	101	5	0.25
		20	1.00

Histogram



Histogram relative frequency



Histograms redrawn from the table above, because Excel's histogram chart type cannot be exported to PDF.